Experiment No. 8

Title

Mounting and Unmounting Devices in Linux

Aim

The problem is to identify storage devices in Linux, mount and unmount file systems manually, and understand the significance of mounting points and device management in system administration and cyber security environments.

Theory

Linux File System and Mounting

Linux follows a single-rooted directory structure where all files and devices are accessible through /. Storage devices must be mounted to a directory (mount point) before their contents can be accessed.

Device Files

Linux represents hardware devices as files located in the /dev directory. Common examples include:

- /dev/sda, /dev/sdb – Hard disks
- /dev/sda1, /dev/sdb1 – Disk partitions

Mount Points

A mount point is an empty directory where a device's file system is attached.

Example:

- /mnt
- /media

Unmounting Devices

Unmounting safely disconnects the device from the file system, ensuring all data is written properly and preventing corruption.

COMMANDS USED-

lsblk	List block devices
fdisk -l	Display disk partition information
mount	Mount a file system
umount	Unmount a file system
df -h	Display mounted file systems
mkdir	Create mount point

Procedure / Steps

A. Identifying Devices Open the terminal.

List available block devices:

lsblk

Display detailed disk information:

sudo fdisk -l

```
┌──(setika㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   50G  0 disk
├─sda1   8:1    0 47.4G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  2.6G  0 part [SWAP]
sr0     11:0    1 1024M  0 rom

┌──(setika㉿kali)-[~]
└─$ sudo fdisk -l
Disk /dev/sda: 50 GiB, 53687091200 bytes, 104857600 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0×d3854adf

Device     Boot    Start       End  Sectors  Size Id Type
/dev/sda1  *        2048  99422207 99420160 47.4G 83 Linux
/dev/sda2       99424254 104855551  5431298  2.6G  f W95 Ext'd (LBA)
/dev/sda5       99424256 104855551  5431296  2.6G 82 Linux swap / Solaris
```

B. Creating a Mount Point

Create a directory to be used as a mount point:

sudo mkdir /mnt/us

C. Mounting a Device

Mount a partition (example /dev/sdb1) to the mount point:

sudo mount /dev/sdb1 /mnt/us

Verify the mount:

df -h mount

```
┌──(setika㉿kali)-[/mnt/us]
└─$ mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=1900052k,nr_inodes=475013,mode=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=401148k,mode=755,inode64)
/dev/sda1 on / type ext4 (rw,relatime,errors=remount-ro)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot,memory_h
tlb_accounting)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=41,pgrp=1,timeout=0,minproto=5,maxproto=5,direc
pipe_ino=6748)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,size=2005736k,nr_inodes=1048576,inode64)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,nosuid,nodev,noexec,relatime)
sunrpc on /run/rpc_pipefs type rpc_pipefs (rw,relatime)
none on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,n
nodes=1024,mode=700,inode64,noswap)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=401144k,nr_inodes=100286,mode=700,uid=1000,gid
00,inode64)
gvfsd-fuse on /run/user/1000/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
none on /run/credentials/getty@tty2.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,n
nodes=1024,mode=700,inode64,noswap)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1
k,nr_inodes=1024,mode=700,inode64,noswap)
```

D. Accessing the Mounted Device

Navigate to the mounted directory:

cd /mnt/usb ls

E. Unmounting the Device

Exit the mounted directory:

cd ~

Unmount the device:

sudo umount /mnt/usb

Verify unmounting:

df -h

Outcomes

- ✓ Storage devices and partitions are listed successfully.
- ✓ Device is mounted to the specified mount point.
- ✓ Files inside the mounted device are accessible.
- ✓ Device is safely unmounted without errors.

Result

- ❖ Storage devices were successfully identified, mounted, accessed, and unmounted in Linux. The student gained practical understanding of Linux file system mounting and device management.